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CNN

For my Convolutional neural network, my architecture contains a convolution layer, an input layer, and flattening. The test loss of my network was 2.307664632797241 which is lower than the loss for the default given neural network. The training loss starts at 2.6 and lowers to 0.23. The val loss starts at 1.9 and has a parabolic relation ending at 1.8 with a low point of 1.6. Lastly, the accuracy was 59%.

All Convolutional Net

The All convolutional network model has several convolution layers, an input layer and flattening. The test loss of this network was 2.167452096939087, lower than my previous CNN. The training loss is constant at 2.2785, and the value loss is constant at 2.1583. Finally, the total test accuracy was 0.18496815860271454 or about 18%.

Regularization

The augmentation I chose involved dropping 30% of the data in the CNN estimator using the Dropout method. The test accuracy from my regularized model is lower than my original CNN with a test accuracy of roughly 31% compared to my original CNN having a test accuracy

of 59%. In addition to this, the regularized model has a starting value loss of 1.869 and a final value loss of 1.621. The regularized model also has a starting training loss of 2.219 and final training loss of 1.564. This training loss is higher than the beginning of training for my original CNN, and much higher compared to my final training loss value for my CNN model. The final Value loss of my regularized model is lower than both peaks in my original CNN, but still higher than the trough in my original CNN, and the initial value loss of my regularized model is higher than both peak losses in my original CNN.

Transfer Learning

The pre-trained model that I chose was the Original CNN that I created as mentioned in the beginning of this report. That model was a convolutional neural network with an overall accuracy of about 59%. The final accuracy of the model was 6.7%. Here is the overall plot for training loss of the model:

